
UNIT 14 DATA WAREHOUSING AND DATA MINING

Structure

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14.0 INTRODUCTION

With the advancement of communication technology, a large amount of data is generated and collected by various organisations. This large data can be used for making meaningful decisions, which can be attributed to discovering useful knowledge and patterns from their existing data. One of the ways of storing vast reservoirs of data is data warehousing. Data warehouses provide superior capabilities for data storage, processing, and responsiveness to analytical queries compared to transaction-oriented databases. In decision-making applications, users need to access a larger volume of data very rapidly – much faster than what can be conveniently handled by traditional database systems. Often, such data is extracted from multiple operational databases. The data of a data warehouse can be used by a data mining application. Data mining is an interdisciplinary field that takes its approach from different areas like databases, statistics, artificial intelligence and data structures to extract hidden knowledge from large volumes of data. The data mining concept is used by the research community and the industry for various kinds of data analysis.

This unit aims to introduce you to some of the fundamental techniques used in data warehousing and data mining. This unit introduces only those data warehousing concepts which relate to structured data.

14.1 OBJECTIVES

After going through this unit, you should be able to:

- Illustrate the concepts of a data warehouse;
- discuss data warehousing schemas;
- identify the multi-dimensional data modelling of a data warehouse;
- explains the purpose of data mining and its applications;
- illustrate classification and clustering approaches of data mining;
- explains how association rules can be identified in data mining.

14.2 WHAT IS DATA WAREHOUSING?

Let us first try to answer the question: What is a data warehouse? A simple answer could be: A data warehouse is a tool that manages data *after and outside* of operational systems. Thus, it is not a replacement for the operational systems but is a major tool that acquires data from the operational systems. Data warehousing technology has evolved in business applications for the process of strategic decision-making. Data warehouses, sometimes, may be considered the key components of the IT strategy and architecture of an organisation.

The formal definition of the data warehouse was given by **W.H. Inman**: “A Data warehouse is a collection of data, which is (i) *subject-oriented*, (ii) *integrated*, (iii) *nonvolatile* (iv) *time-variant* that supports decision-making by the management”. *Figure 1* presents some uses of data warehousing in various industries.

S.No.	Industry	Functional Areas of Use	Strategic Uses
1	Banking	Creating new schemes for loans and other banking products; helps in operations; identifies information for marketing.	Finding trends for customer service; product and service promotions; reduction of expenses.
2	Airline	Operations; marketing.	Crew assignment; aircraft maintenance plans; fare determination; analysis of route profitability; frequent - flyer program design.
3	Hospital	Operation optimisation.	Reduction of operational expenses; scheduling of resources.
4	Investment and Insurance	Insurance product development; marketing	Risk management; financial market analysis; customer tendencies analysis; portfolio management

Figure 1: Uses of Data Warehousing

A data warehouse offers the following advantages:

- It provides historical information that can be used in many different forms of analysis.
- Improves the data quality by predicting missing data.
- It can help support disaster recovery, although not alone, but with other backup resources.

One of the major advantages of a data warehouse is that it allows a large collection of historical data from many operational databases that can be analysed through one data warehouse interface. A data warehouse does not create value of its own in an organisation. However, the value can be generated by the users of the data warehouse. For example, a data warehouse of a manufacturing unit may answer some of the following questions:

- What would be the income, expenses and profit for a year?
- What would be the sales amount this month?
- Who are the vendors for a product that is to be procured?
- How much of each product is manufactured in each production unit?
- How much is to be manufactured?

- What percentage of the product is defective?
- Are customers satisfied with the quality? etc.

The data warehouse supports various business intelligence applications. Some of these may be - online analytical processing (OLAP), decision-support systems (DSS), data mining etc. We shall discuss some of these terms in more detail in the later sections.

14.3 BASIC COMPONENTS OF A DATA WAREHOUSE

A data warehouse is defined as a *subject-oriented, integrated, nonvolatile, time-variant collection*, but how can we achieve such a collection? To answer this question, let us define the basic architecture that helps a data warehouse achieve the objectives stated above. We shall also discuss various processes that are performed by these components on the data.

Figure 2 defines the basic architecture of a data warehouse. The analytical reports are not a part of the data warehouse but are one of the major business application areas including OLAP, DSS and Data Mining

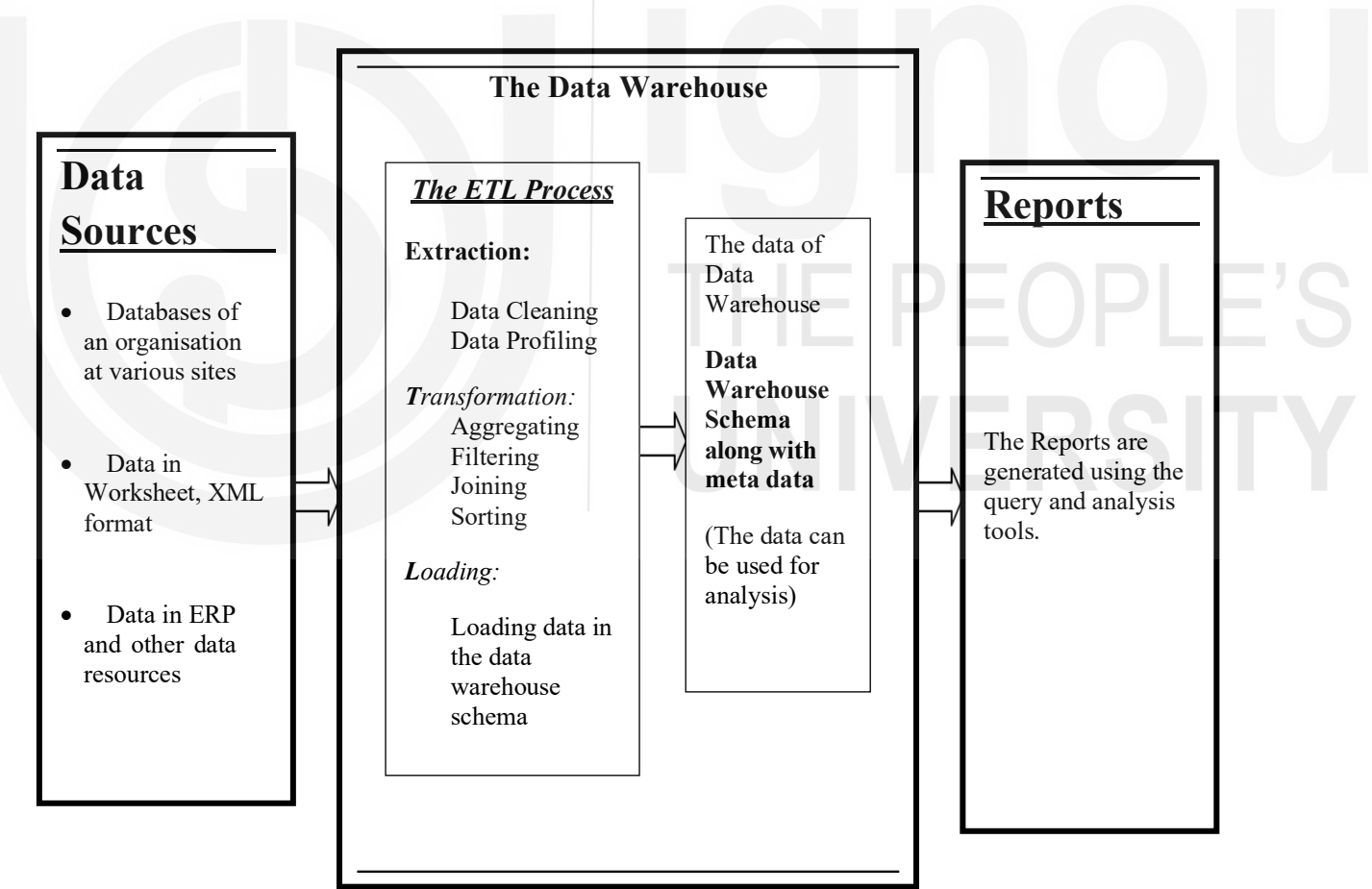


Figure 2: The Data Warehouse Architecture

14.3.1 The Data Sources

The data of the data warehouse can be obtained from many operational systems. A data warehouse interacts with the environment that provides most of the source data for the data warehouse. By the term environment, we mean traditionally developed database

systems and other applications. In a large installation, hundreds or even thousands of these database systems or files-based systems exist with plenty of redundant data.

The warehouse database obtains most of its data from such different forms of legacy systems – files and databases. Data may also be sourced from external sources as well as other organisational systems. This data needs to be integrated into the warehouse. But how do you integrate the data of these large numbers of operational systems into the data warehouse system? You need the help of ETL tools to do so. These tools capture the data that is required to be put in the data warehouse.

Data in Data Warehouse

The basic characteristics of the data in a data warehouse can be described in the following way:

- i) **Subject Oriented:** The first characteristic of the data warehouse's data is that its design and structure can be oriented to important objects of the organisation. These objects for a university data warehouse can be STUDENT, PROGRAMME, REGIONAL CENTRES etc., whereas the operational systems may be designed around applications and functions such as ADMISSION, EXAMINATION and RESULT DECLARATIONS (in the case of a University). Refer to *Figure 3*.

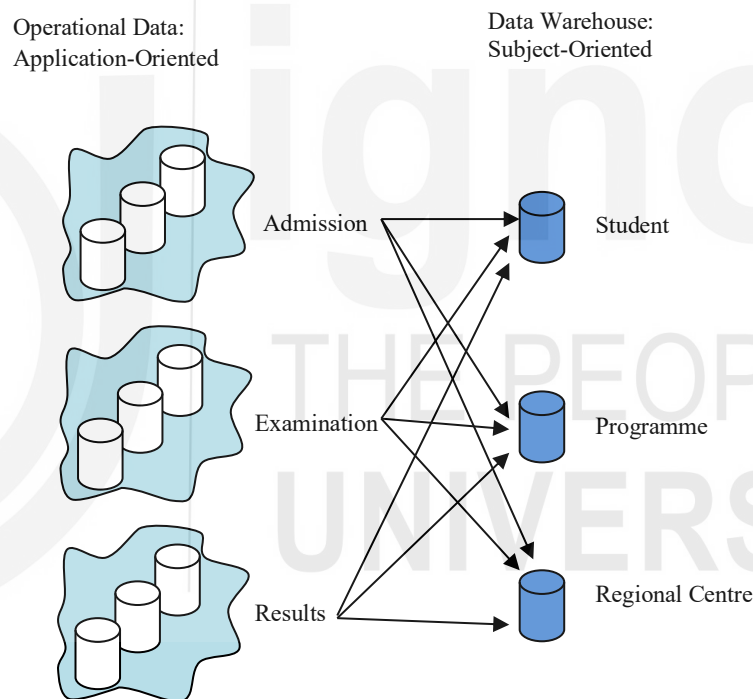


Figure 3: Operations system data vs Data warehouse data

- ii) **Integrated:** Integration means bringing together data from multiple dissimilar operational sources on the basis of an enterprise data model. A data model of a data warehouse can be a basic template that uniquely defines the organisation's key data items. Data integration requires:

- Standardising the data naming and definition: for example, “student enrolment number” can be standardised over the data of the entire university.
- Standardising of encodings: for example, the first two digits of an enrolment number represent the year of admission.
- Standardising the Measurement of variables: for example, all the units would be expressed in the metric system and all monetary details would be given in Indian Rupees.

- iii) **Time-Variant:** The third defining characteristic of the data in a data warehouse is that it is time-variant or historical in nature. The entire data in the data warehouse is/was accurate at some point in time. This is in contrast with operational data that changes over a shorter time period. *Figure 4* defines this characteristic of a data warehouse.

OPERATIONAL DATA	DATA WAREHOUSE DATA
It is the current value data	Contains a snapshot of historical data
Time span of data = 60-90 days	Time span of data = 5-10 years or more
Data can be updated in most cases	After making a snapshot of the data record cannot be updated
May or may not have a timestamp	Will always have a timestamp

Figure 4: Characteristics of data of operational data and data in a data warehouse

- iv) **Non-volatile:** Data loaded in a data warehouse is subsequently scanned and used but is not updated in the same classical ways as the operational system's data, which is updated through the transaction processing cycles.

Metadata Directory

The metadata directory component defines the repository of the information stored in the data warehouse. The metadata can be used by general users as well as data administrators. It contains the following information:

- i) the structure of the contents of the data warehouse data,
- ii) the source of the data,
- iii) the data transformation processing requirements, such that data can be passed from the legacy systems into the data warehouse database,
- iv) the process summarisation of data,
- v) the data extraction history, and
- vi) how the data needs to be extracted from the data warehouse.

14.3.2 Data Extraction, Transformation and Loading (ETL)

The first step in data warehousing is to perform data extraction, transformation, and loading of data into the data warehouse. This is called ETL, which is Extraction, Transformation, and Loading. ETL refers to the methods involved in accessing and manipulating data available in various sources and loading it into a target data warehouse.

What happens during the ETL Process?

The following are the sub-processes of the ETL process of the data warehouse:

Data Extraction: The ETL is a three-stage process. During the *Extraction* phase, the desired data is identified and extracted from many different sources. These sources may be different databases or non-databases. The process of extraction sometimes involves some basic transformation. For example, if the data is being extracted from two Sales databases where the sales in one of the databases are in Dollars and the other in Rupees, then a simple transformation would be required in the data. The size of the extracted data may vary from hundreds of kilobytes to hundreds of gigabytes, depending on the data sources and business systems.

The *extraction* process involves data cleansing and data profiling. Data cleansing can be defined as the process of removal of inconsistencies in the data. For example, the

state name may be written in many ways, and they can be misspelt too. For example, the state Uttar Pradesh may be written as U.P., UP, Uttar Pradesh, Utter Pradesh etc. The cleansing process may try to correct the spellings as well as resolve such inconsistencies. But how does the cleansing process do that? One simple way may be to create a Database of the States with possible fuzzy matching algorithms that may map various variants into one state name. Thus cleansing the data to a great extent.

Data profiling involves creating the necessary data from the point of view of a data warehouse application. Another concern here is to eliminate duplicate data. For example, an address list collected from different sources may be merged and purged to create an address profile with no duplicate data.

Data Transformation: One of the most time-consuming tasks - data *transformation* follows the extraction stage. The data transformation process includes the following:

- Use of data filters,
- Data validation against the existing data,
- Checking data duplication, and
- Information aggregation.

Transformations are useful for transforming the source data according to the requirements of the data warehouse. The process of transformation should ensure the quality of the data that needs to be loaded into the target data warehouse. Some of the common transformations are:

Filter Transformation: Filter transformations are used to filter the rows in a mapping that do not meet specific conditions. For example, the list of employees of the Sales department who made sales above Rs.50,000/- may be filtered out.

Joiner Transformation: This transformation is used to join the data of one or more different tables that may be stored in two different locations and could belong to two different sources of data that may be relational or other data formats like XML.

Aggregator Transformation: Such transformations perform aggregate calculations on the extracted data. Some such calculations may find the sum or average.

Sorting Transformation: requires creating an order in the required data based on the application requirements of the data warehouse.

Data Loading: Once the data for the data warehouse is properly extracted and transformed, it is *loaded* into a data warehouse. This process requires creation and execution of programs that perform this task. One of the key concerns here is to propagate source data updates. Sometimes, this problem is equated to the problem of maintenance of the materialised views.

When should we perform the ETL process for the data warehouse? ETL process should normally be performed at night or when the load on the operational systems is low. **Please note** that the integrity of the extracted data can be ensured by synchronising different operational applications feeding the data warehouse and the data in the data warehouse.

Check Your Progress – 1

1) What is a Data Warehouse?

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2) What are the characteristics of data in a data warehouse?

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3) What is ETL? What are the different transformations that are needed during the ETL process?

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14.4 MULTIDIMENSIONAL DATA MODEL OF A DATA WAREHOUSE

A data warehouse is a huge collection of data. Such data may involve grouping of data on multiple attributes. For example, the enrolment data of the students at a University may be represented using a student schema such as:

Student_enrolment (year, programme, region, number)

Some data values for such schema are (Also refer to *Figure 5*, which shows this data):

- In the year 2002, BCA enrolment at Region (Regional Centre Code) RC-07 (Delhi) was 350.
- In the year 2003, BCA enrolment in Region RC-07 was 500.
- In the year 2002, MCA enrolment in all the regions was 8000.

Please note that for representing the value of a number of students, you need to refer to three attributes: year, programme and region. Each of these attributes is identified as a dimension attribute. Thus, the data of the *Student_enrolment* table can be modelled using three-dimensional attributes (year, programme, region) and a measure attribute (number). Such kind of data is referred to as multidimensional data. Thus, a data warehouse may use multidimensional matrices referred to as a data cube model. If the dimensions of the matrix are greater than three, then it is called a hypercube. Figure 5 represents the multidimensional data of a university:

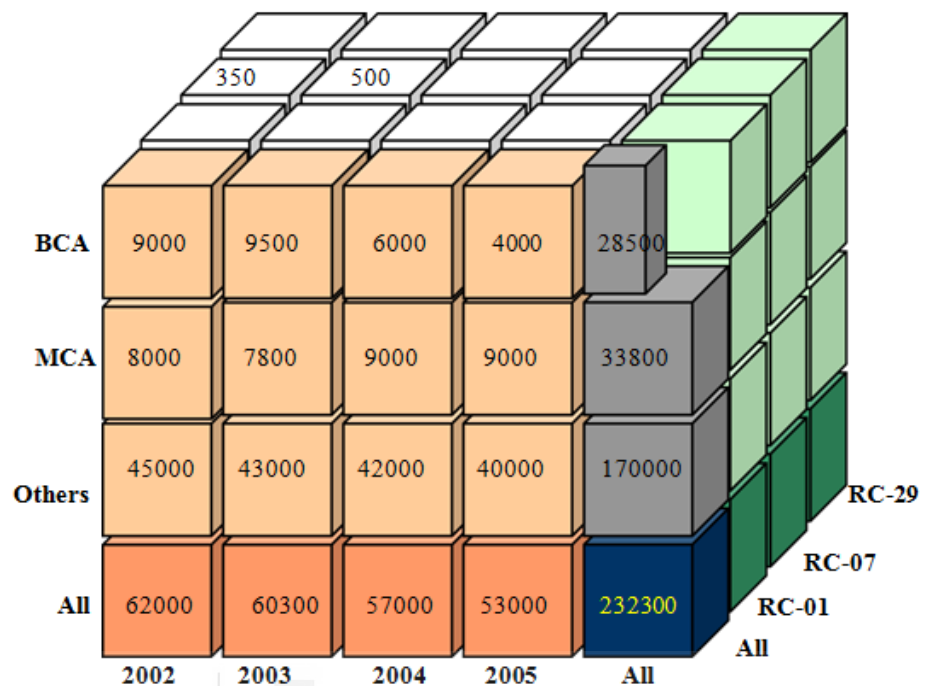


Figure 5: Sample multidimensional data

Multidimensional data may be a little difficult to analyse. Therefore, Multidimensional data may be displayed on a certain pivot, for example, consider the following table:

Region: ALL THE REGIONS				
	BCA	MCA	Others	All the Programmes
2002	9000	8000	45000	62000
2003	9500	7800	43000	60300
2004	6000	9000	42000	57000
2005	4000	9000	40000	53000
ALL the Years	28500	33800	170000	232300

Figure 6: Pivot table on Programme and Year dimension.

The table given above shows the multidimensional data in *cross-tabulation*. This is also referred to as a *pivot table*. Please note that cross-tabulation is a two-dimensional structure. Therefore, if the data warehouse has three dimensions, then the cross-tabulation will be done on only two dimensions and the third dimension would be kept as ALL. For example, the table above has two dimensions Year and Programme, the third dimension Region has a fixed value ALL for the given table. The last row of Figure 6 is on one dimension, viz. Programme.

Now, the question is, how can multidimensional data be represented in a data warehouse? or, more formally, what is the schema for multidimensional data?

Three common multidimensional schemas are the Star schema, the Snowflake schema and Galaxy or Fact Constellation schema. In this Unit, we will define the Star and Snowflake schema. Galaxy schema contains multiple fact tables. You can find more detail on this schema in the further readings. A multidimensional storage model contains two types of tables: the dimension tables and the fact table. The dimension tables have tuples of dimension attributes, whereas the fact tables have one tuple each for a recorded fact. Let us demonstrate this with the help of an example. Consider the University data warehouse where one of the data tables is the *Student_enrolment* table. The three dimensions in such a case would be:

- Year
- Programme, and
- Region

The star schema for such a data is shown in *Figure 7*.

Dimension Table:
Programme

ProgramCode
Name
Duration

Fact Table:
Enrolment

Year
Programme
Region
Enrolment

Dimension Table:
Table: Year

Year
Semester
Start date

Dimension Table:
Region

RCcode
RCname
City
State
RCPhone

Figure 7: A Star Schema

Please note that in *Figure 7*, the fact table points to different dimension tables, thus, ensuring the reliability of the data. Please also note that each Dimension table is a table for a single dimension only and that is why this schema is known as a Star schema. However, a dimension table may not be normalised. Thus, a new schema named the Snowflake schema was created. A Snowflake schema has normalised but hierarchical dimensional tables. For example, consider the Star schema shown in *Figure 7*, if, in the Region dimension table, the value of the field Rcphone is multivalued, then the Region dimension table is not normalised. Thus, we can create a Snowflake schema for such a situation (Refer to Figure 8).

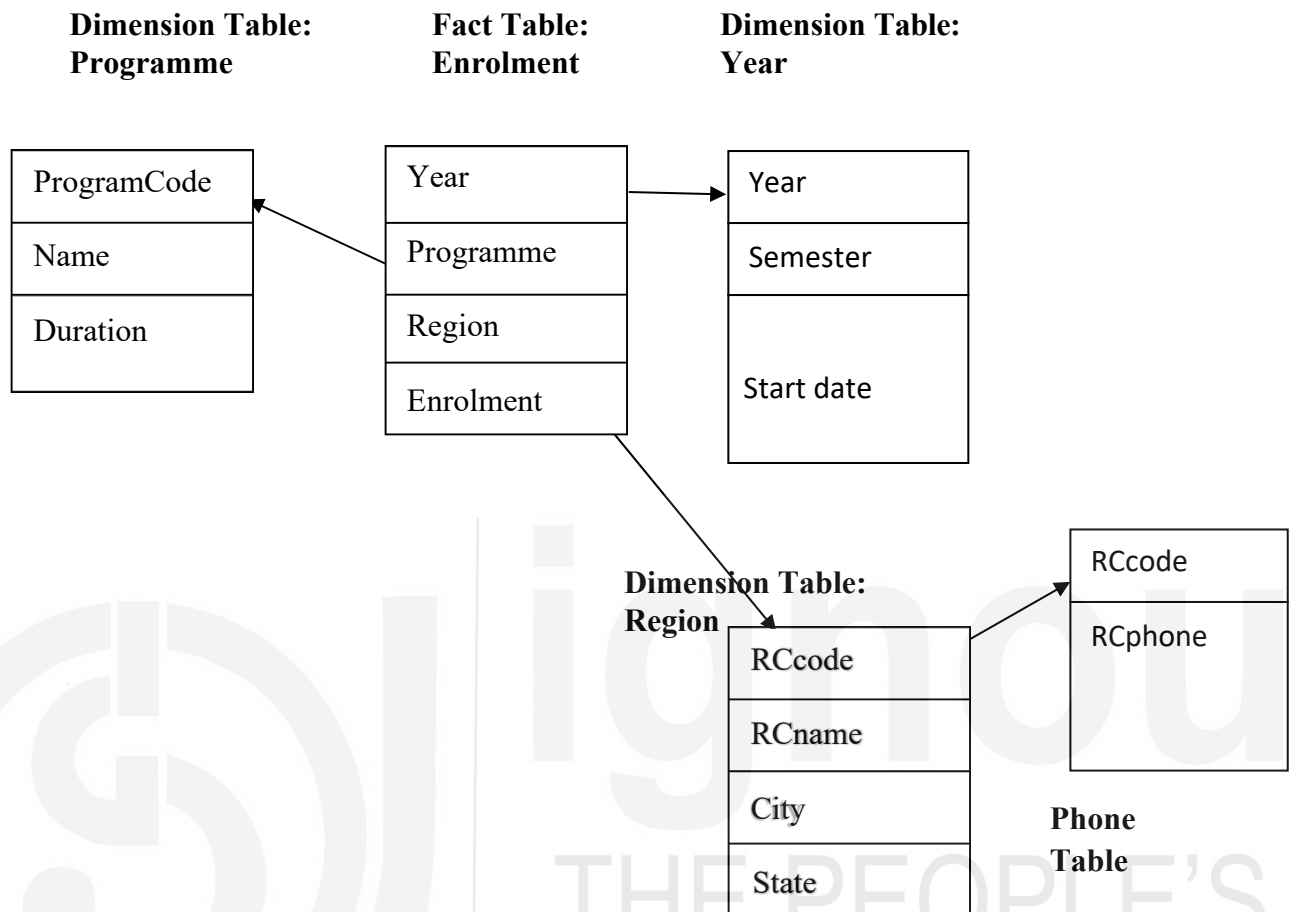


Figure 8: Snowflake Schema

Data warehouse storage can also utilise indexing to support high-performance access. As the data of a data warehouse is non-volatile, the data warehouse allows storage and access to summary data.

A data warehouse is an integrated collection of data and can help the process of making better business decisions. Several tools and methods are available to process the data of a data warehouse to create information and knowledge that supports business decisions. Two such techniques are Decision-support systems and Online Analytical processing. A detailed discussion on these topics is beyond the scope of this unit. In this unit, we define data mining, which can extract useful information from a data warehouse.

14.5 DATA MINING TECHNOLOGY

Data is growing at a phenomenal rate today and users expect more sophisticated information from data. There is a need for techniques and tools that can automatically generate useful information and knowledge from large volumes of data. Data mining is one such technique of generating hidden information from the large data.

Data mining can be defined as: “an automatic process of extraction of non-trivial or implicit or previously unknown but potentially useful information or patterns from data in large databases, data warehouses or in flat files”.

Data mining uses the data of a data warehouse, which is well equipped for providing data as input for data mining. The advantages of using the data of a data warehouse for data mining are listed below:

- Data quality and consistency are essential for data mining to ensure the accuracy of the predictive models. Data is loaded in a data warehouse after data extraction, cleaning and transformation; therefore, is good quality data.
- As stated earlier, the data of a data warehouse is integrated from multiple data sources for a specific purpose and, therefore, is suited for data mining.
- Some data mining techniques would require aggregated or summarised data. A data warehouse contains such data.

As defined earlier, data mining generates potentially useful information or patterns from data. In fact, the information generated through data mining can be used to create knowledge. So let us first define the three terms: data, information and knowledge.

1. **Data (Symbols)** are raw facts, for example, “Miral” is a name, BCA is a programme, and 2200105000 is an enrolment number.
2. **Information:** Information is the processed data. It provides answers to “who”, “what”, “where”, and “when” questions. For example, Miral is a BCA student of IGNOU having enrolment number “2200105000”. He has scored 761 marks out of 1000.
3. **Knowledge:** Knowledge is the application of data and information, it answers the “how” questions. This is not explicit in the database. A student who scores more than 075% marks is brilliant; therefore, Miral is brilliant.

Data Mining Approaches

The approaches to data mining are based on the type of information/ knowledge to be mined. We will emphasise three different approaches: Classification, Clustering, and Association Rules.

The classification task maps data tuples into predefined groups or classes. The task of clustering is to group tuples with similar attribute values into a new class. So, classification is supervised by the goal attribute, while clustering is an unsupervised classification. The task of association rule mining is to find relationships between/among data values of a set of transactional data. Its original application was on “market basket data”.

In most of these approaches, a notion of distance measure is used. A distance measure is used to find the distance or dissimilarity between objects. The two most common distance measures are:

- Euclidean distance:
$$\text{edis}(t_i, t_j) = \sqrt{\sum_{h=1}^k (t_{ih} - t_{jh})^2}$$
- Manhattan distance:
$$\text{mdis}(t_i, t_j) = \sum_{h=1}^k |(t_{ih} - t_{jh})|$$

where t_i and t_j are tuples and h can take the values from 1 to k , each of which represents a different attribute that is being used to compute the distance.

In the next section, we discuss the three data mining approaches.

Check Your Progress – 2

1) What is a dimension? How is it different from a fact table?

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2) How is the Snowflake schema different from the Star schema?

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3) Define what data mining is.

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4) What are different data mining tasks?

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14.6 CLASSIFICATION

The classification task maps data tuples into predefined groups or classes.

Given a database/dataset consisting of tuples t_i , where i varies from 1 to n , i.e.

$D = \{t_1, t_2, \dots, t_n\}$;

and a set of known classes $C = \{C_1, \dots, C_m\}$, where $m \gg n$.

The classification problem is to map each t_i to a C_i .

Some simple examples of classification are:

- Teachers classify students' marks data into a set of grades as A, B, C, D, or F.
- You can classify the height of a set of students in the classes: tall, medium or short.

Basic Principle:

1. The classification involves learning using the training data, which is the data that has already been assigned to one of the classes. This training results in a classification model.
2. This model is then tested using the test data to find the effectiveness of the model. The test data also has assigned classes, which are checked against the class predicted by the model. The accuracy of the model can be ascertained on the basis of correctly predicted classes of the test data.
3. A good model is then used to classify the data that has not been classified.

Some of the common techniques used for classification are Decision Trees, Neural Networks etc. In this section, we present two examples of classification.

14.6.1 Classification Using Distance (K-Nearest Neighbour)

This approach places items in the class to which they are “closest” by determining the distance of an item from a class. Classes are represented by a central point called centroid. The K-nearest neighbour algorithm has the following steps:

- 1) Create a training data set consisting of attributes or features that would be used for classifying data and the identified classes for these attributes. Figure 9 shows an example training data set.
- 2) Defines the number of *near items* (items that have less distance to the attributes of concern) from the training data that should be used to classify data. The value of K should be $\leq \sqrt{\text{Number_of_Training_Items}}$
- 3) A new item is placed in the class in which most of its *near items* are placed.

Example: Consider the following data, which classifies each person’s class <Short, Medium, Tall> depending upon height attribute.

Name	Height	Class
Sunita	1.6m	Short
Ram	2.0m	Tall
Namita	1.9m	Medium
Radha	1.88m	Medium
Jully	1.7m	Short
Arun	1.85m	Medium
Shelly	1.6m	Short
Avinash	1.7m	Short
Sachin	2.2m	Tall
Manoj	2.1m	Tall
Sangeeta	1.8m	Medium
Anirban	1.95m	Medium
Krishna	1.9m	Medium
Kavita	1.8m	Medium
Pooja	1.75m	Medium

Figure 9: Sample Height data with classification

- 1) You are given the task of classifying the tuple xyz <XYZ, 1.6> using the data that is given to you.
- 2) The *height* attribute is used for distance calculation, and suppose K=5, then the following are the five nearest tuples to the tuple xyz. Please note that we have used the Manhattan distance on attribute *height* as a measure of classification.

Height	Class
1.6m	Short
1.7m	Short
1.6m	Short
1.7m	Short
1.75m	Medium

- 3) On examination of the tuples above, we classify the tuple xyz<XYZ, 1.6> to the *Short* class since most of the nearest tuples belong to the *Short* class.

Thus, in K-nearest neighbour classification, the classification is controlled by the neighbours.

14.6.2 Decision Tree

Given a data set $D = \{t_1, t_2, \dots, t_n\}$, where tuple $t_i = \langle A_1, A_2, \dots, A_h \rangle$, that is, each tuple is represented by h attributes. Also, let us suppose that the classes are $C = \{C_1, \dots, C_m\}$, then the Decision or Classification Tree is a tree associated with D such that:

- The tree consists of internal and leaf nodes. A label using an attribute A_i is assigned to each internal node.
- An arc from a parent node is labelled with a predicate on the attribute label of the parent node.
- Every leaf node has a class label.

The basic steps in the Decision Tree are as follows:

- Building the tree by using the training set dataset/database.
- Applying the tree to the new dataset/database.

This decision tree can then be used for classification. We show an example of a decision tree without giving an algorithm for drawing it. You may refer to further reading for details.

Consider the following data in which the *Position* attribute acts as a predicted class, and department and salary are attributes of determining the class of a tuple.

Department	Age	Salary	Position
Personnel	31-40	Medium Range	Boss
Personnel	21-30	Low Range	Assistant
Personnel	31-40	Low Range	Assistant
MIS	21-30	Medium Range	Assistant
MIS	31-40	High Range	Boss
MIS	21-30	Medium Range	Assistant
MIS	41-50	High Range	Boss
Administration	31-40	Medium Range	Boss
Administration	31-40	Medium Range	Assistant
Security	41-50	Medium Range	Boss
Security	21-30	Low Range	Assistant

Figure 10: Sample data for classification

You may analyse the data given in Figure 10 on each individual attribute. The following tables present the analysis of class Position on the attributes Age, Department and Salary, respectively.

Age	Assistant	BOSS
21-30	4	0
31-40	2	3
41-50	0	2

Department	Assistant	BOSS
Personal	2	1
MIS	2	2
Administration	1	1
Security	1	1

Salary	Assistant	BOSS
Low Range	3	0
Medium Range	3	3
High Range	0	2

Out of the three attributes, the Age attribute predicts the Position class the best, as it predicts that all the person in the range 21-30 are Assistants and all the person in the age group 41-50 are Boss. Thus, it should be used as the first splitting attribute. However, for a person in the age range of 31-40, the Position cannot be defined. So, we

have to find the splitting attribute for this age range 31-40. The tuples that belong to this range are as follows:

Tuples Only for 31-40 age range

Department	Salary	Position
Personnel	Medium Range	Boss
Personnel	Low Range	Assistant
MIS	High Range	Boss
Administration	Medium Range	Boss
Administration	Medium Range	Assistant

The tuples as given above can be mapped as:

Department	Assistant	BOSS
Personal	1	1
MIS	0	1
Administration	1	1

Salary	Assistant	BOSS
Low Range	1	0
Medium Range	1	2
High Range	0	1

The Salary attribute can perform better prediction than the Department attribute, so we select Salary as the next splitting attribute. In the middle range Salary, the Position class is not defined while for other ranges it is defined. So, we have to find the splitting attribute for this middle range. Since only Department attribute is left, so, Department will be the next splitting attribute. Now, the tuples that belong to this salary range are as follows:

Department	Position
Personnel	Boss
Administration	Boss
Administration	Assistant

In the Personnel department, all person are Boss, while, in the Administration there is a tie between the classes. So, the person can be either Boss or Assistant in the Administration department. The final decision tree, based on presented data, for predicting the Position class is as follows:

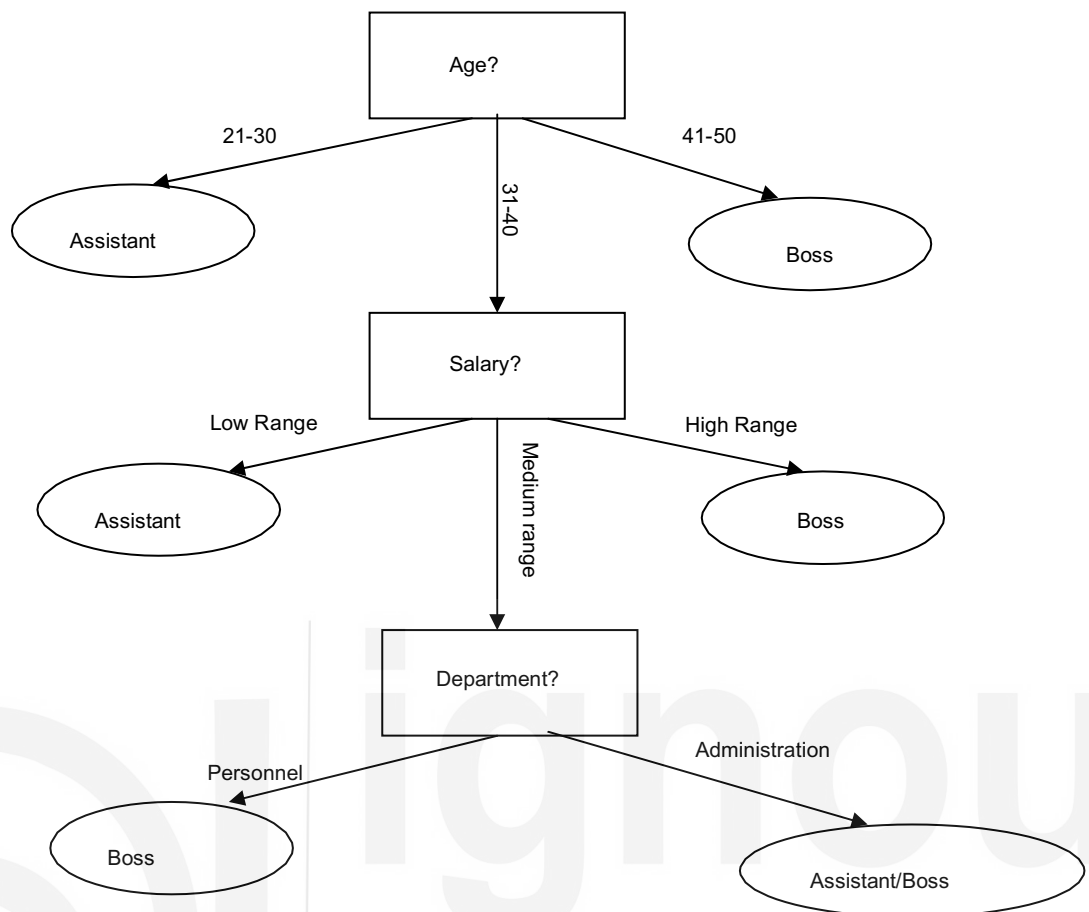


Figure 11: The decision tree for the sample data of Figure 10

Now, you can use the decision tree to predict the Position of a person if his/her Age, Salary and Department attributes are known.

There are many more techniques of classification, however, they are beyond the scope of this unit, you may refer to further readings for these.

14.7 CLUSTERING

Clustering is grouping tuples with similar attribute values into the same group. Given a database $D = \{t_1, t_2, \dots, t_n\}$ of tuples and an integer value K , the Clustering problem is to define a mapping where each tuple t_i is assigned to one cluster K_j , $1 \leq j \leq K$.

Unlike the classification problem, clusters are not known in advance. The user has to enter the value of the number of clusters K . In other words, a *cluster* can be defined as the collection of data objects that are similar in nature, as per certain defining properties, but these objects are dissimilar to the objects in other clusters. Clustering is a very useful exercise, especially for identifying similar groups in the given data. Such data can be about buying patterns, geographical locations, web information and many more. For example, you can use clustering to segment the customer database of a departmental store based on similar buying patterns. Similarly, to identify similar Web usage patterns, you may use clustering. Some of the clustering issues are as follows:

- **Outlier handling:** How will the outliers be handled? (Outliers are data objects that have values far beyond the average data limits of those data objects). Outlier handling requires answering the question: Whether an outlier be

considered or left aside while calculating the clusters?

- **Dynamic data:** How will you handle dynamic data?
- **Interpreting results:** How will the result be interpreted?
- **Evaluating results:** How will the result be calculated?
- **Number of clusters:** How many clusters will you consider for the given data?
- **Data to be used:** whether you are dealing with quality data or noisy data? If the data is noisy, how is it to be handled?
- **Scalability:** Whether the algorithm that is used is to be scaled for small as well as large data sets/databases.

There are many algorithms for clustering. However, we will discuss only one basic algorithm. You can refer to more details on clustering from further readings.

Partitioning Clustering

The partitioning clustering algorithm constructs K partitions from a given set of n objects of data. Here, $K \leq n$, and each partition must have at least one data object while one object belongs to **only one** of the partitions. A partitioning clustering algorithm normally requires users to input the desired number of clusters, K . We define one such technique called K -means clustering with the help of an example.

K -Means clustering: In K -Means clustering, initially a set of K clusters is randomly chosen. Then iteratively, items are moved among sets of clusters until the desired set of clusters is reached. A high degree of similarity among elements in a cluster is obtained by using this algorithm. For this algorithm, a set of clusters $K_i = \{t_{i1}, t_{i2}, \dots, t_{im}\}$ is chosen. A cluster mean is computed for each cluster using the equation:

$$m_i = (1/m) (t_{i1} + \dots + t_{im}) \quad \dots\dots\dots(1)$$

Where t_i represents the tuples and m represents the mean. This mean is used to refine the clusters further. We explain this algorithm with the help of an example.

Example: Consider an input set is given as:

Input set= {1, 2, 3, 5, 10, 15, 16, 18, 22, 30} and $K = 2$.

Find the cluster of input values.

Step 1: Randomly assign means to two clusters, K_1 and K_2 , as $m_1=3$ and $m_2=5$

Assign Input set values to clusters based on the smallest distance to the mean.

This will result in:

$$K_1 = \{1, 2, 3\} \text{ and } K_2 = \{5, 10, 15, 16, 18, 22, 30\}$$

Step 2: Compute the mean (use the formula given in equation 1) of the clusters after the assignment of the input set. This will result in $m_1=2$ and $m_2=16.57$.

Redefine clusters as per the closest mean from the mean of the cluster. For example, for the mean of step 2, input value 5 is closer to 2, so add it to K_1 . Thus, the revised clusters are:

$$K_1 = \{1, 2, 3, 5\} \text{ and } K_2 = \{10, 15, 16, 18, 22, 30\}.$$

Step 3: Compute the mean for the latest cluster assignment. This will result in $m_1=2.75$ and $m_2=18.5$.

Repeat the process of finding the cluster as per the closest mean from the mean of the cluster. The revised clusters remain as:

$$K_1 = \{1, 2, 3, 5, 10\} \text{ and } K_2 = \{15, 16, 18, 22, 30\}.$$

Step 4: Compute the mean, it will be $m_1=4.2$, $m_2=20.2$

No change in clusters, so STOP. The final clusters are:

$$K_1 = \{1, 2, 3, 5, 10\} \text{ and } K_2 = \{15, 16, 18, 22, 30\}.$$

You can refer to more details on the clustering algorithms from further readings.

Check Your Progress – 3

1) What is the classification of data? Give some examples of classification.

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2) What is clustering?

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3) How is clustering different from classification?

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14.8 ASSOCIATION RULE MINING

The task of association rule mining is to find certain association relationships among a set of items in a dataset/database. The association relationships are described as association rules. In association rule mining there are two measurements, *support* and *confidence*. The confidence measure indicates the rule's strength, while support corresponds to the frequency of the pattern.

A typical example of an association rule created by data mining often termed “market basket data” is “While shipping for basic groceries, it has been found that about 80% of the customers who purchase bread from the store also purchase butter along with.”

Applications of association rule mining include cache customisation, advertisement personalisation, store layout designing, customer segmentation etc. All these applications try to determine the associations between data items, if it exists, to optimise performance.

Formal Definition:

Given a set of items $I = \{i_1, i_2, \dots, i_m\}$ and a set of transactions T , where a transaction T_i comprises of items and is a subset I . Each transaction is identified by a TID. For the given sets, an association rule is defined as $X \rightarrow Y$, where both X and Y are the subsets of I such that $X \cap Y = \emptyset$.

The support (s) for the stated association rule $X \rightarrow Y$ is defined as the percentage of transactions in T that contains $X \cup Y$.

The confidence (c) for the stated association rule $X \rightarrow Y$ is defined as the percentage of transactions that include X , also include Y .

Support indicates how frequently the pattern occurs, while confidence indicates the strength of the rule.

The objective of association rule mining is to find all those rules which have better support than the defined minimum support and better confidence than the defined

minimum confidence. The association rule mining consists of two sub-problems:

- (1) Find the frequent item sets (*FreqItem*) having support more than a predetermined minimum support.
- (2) Derive association rules from *FreqItem*, which have confidence more than the minimum confidence.

There are a lot of ways to find the large item sets, but in this unit, we will only discuss the Apriori Algorithm.

Apriori Algorithm: For finding frequent item sets.

The Apriori algorithm applies the concept that all the subsets of a frequent itemset should be frequent. The Apriori algorithm generates the candidate item sets from the item sets that were found to be frequent in the previous iteration of the algorithm.

This algorithm first finds the frequent item sets having just 1 item, which are combined to form item sets of 2 items and so on. The algorithm iterations terminates when no additional higher item sets can be generated.

Notations that are used in the Apriori algorithm are given below:

k-itemset	An itemset having k items
L_k	Set of frequent k-itemset (those with minimum support)
C_k	Set of candidates k-itemset (for finding frequent item sets)

The Apriori algorithm is implemented as an iterative function. It takes L_{k-1} as an input parameter and returns L_k . It consists of a join step and a pruning step. It is explained with the help of the following example:

Example: Finding frequent item sets:

Consider the following transactions and find the frequent item sets by applying the Apriori algorithm assuming minimum support (s) =30%.

Transaction ID	Item(s) purchased
1	Shirt, Trouser
2	Shirt, Trouser, Coat
3	Coat, Tie, Tiepin
4	Coat, Shirt, Tie, Trouser
5	Trouser, Belt
6	Coat, Tiepin, Trouser
7	Coat, Tie
8	Shirt
9	Shirt, Coat
10	Shirt, Handkerchief

Figure 12: Sample Transactions data

The method of finding the frequent itemset is shown in the Figure 13.

Iteration Number	Candidates	Frequent itemset ($s \geq 3$, i.e. 30% of 10 Transactions)
1	$C_1 = \{ \text{Belt} \quad 1, \text{Coat} \quad 6, \text{Handkerchief} \quad 1, \text{Shirt} \quad 6, \text{Tie} \quad 3, \text{Tiepin} \quad 2, \text{Trouser} \quad 5 \}$	$L_1 = \{ \text{Coat} \quad 6, \text{Shirt} \quad 6, \text{Tie} \quad 3, \text{Trouser} \quad 5 \}$
2	$C_2 = \{ \{ \text{Coat, Shirt} \} \quad 3, \{ \text{Coat, Tie} \} \quad 3, \{ \text{Coat, Trouser} \} \quad 3, \{ \text{Shirt, Tie} \} \quad 1, \{ \text{Shirt, Trouser} \} \quad 3, \{ \text{Tie, Trouser} \} \quad 1 \}$	$L_2 = \{ \{ \text{Coat, Shirt} \} \quad 3, \{ \text{Coat, Tie} \} \quad 3, \{ \text{Coat, Trouser} \} \quad 3, \{ \text{Shirt, Trouser} \} \quad 3 \}$
3	$C_3 = \{ \{ \text{Coat, Shirt, Trouser} \} \quad 2 \}$	$L_3 = \emptyset$

Figure 13: Frequent Itemset for different k using Apriori algorithm

In pass number 1, you may notice that Belt is purchased in only one (5th transaction) of all the 10 transactions that is why it has a value 1. Similarly, the support of each item is computed from the transaction data.

Likewise, in pass 2 {Coat, shirt} together are purchased in transactions 2, 4, 9.

Also please note that set C_2 is computed by joining set L_1 with itself, which will result in

$C_2 = \{ \{ \text{Coat, Shirt} \}, \{ \text{Coat, Tie} \}, \{ \text{Coat, Trouser} \}, \{ \text{Shirt, Tie} \}, \{ \text{Shirt, Trouser} \}, \{ \text{Tie, Trouser} \} \}$

Set L_2 is computed after finding the support of each element of set C_2 as shown in Figure 13.

The calculation of 3-itemsets is mentioned below:

Join operation on L_2 onto itself yields 3 item sets as:

$\{ \{ \text{Coat, Shirt, Tie} \}, \{ \text{Coat, Shirt, Trouser} \}, \{ \text{Coat, Tie, Trouser} \} \}$

However, the Prune operation removes two of these items from the candidate set C_3 due to the following reasons:

- {Coat, Shirt, Tie} is pruned as one of its subset {Shirt, Tie} is not in L_2
- {Coat, Shirt, Trouser} is retained as {Coat, Shirt}, {Coat, Trouser} and {Shirt, Trouser} all three are in L_2
- {Coat, Tie, Trouser} is pruned as its subset {Tie, Trouser} is not in L_2

The algorithm terminates, as L_3 is a NULL set. Thus, the frequent 2 items are:

$L_2 = \{ \{ \text{Coat, Shirt} \}, \{ \text{Coat, Tie} \}, \{ \text{Coat, Trouser} \}, \{ \text{Shirt, Trouser} \} \}$

Finding Association Rules: Assuming minimum confidence (c) of 60%.

Confidence for rule Coat $\rightarrow$ Shirt

$$= \frac{\text{Transactions containing } \{ \text{Coat, Shirt} \}}{\text{Transactions containing only } \{ \text{Coat} \}} \times 100$$

$$= \frac{3}{6} \times 100 = 50\%$$

Confidence for rule Shirt $\rightarrow$ Coat $= \frac{3}{6} \times 100 = 50\%$

Confidence for rule Coat $\rightarrow$ Tie $= \frac{3}{6} \times 100 = 50\%$

Confidence for rule Tie $\rightarrow$ Coat $= \frac{3}{3} \times 100 = 100\%$

$$\text{Confidence for rule Coat} \rightarrow \text{Trouser} = \frac{3}{6} \times 100 = 50\%$$

$$\text{Confidence for rule Trouser} \rightarrow \text{Coat} = \frac{3}{5} \times 100 = 60\%$$

$$\text{Confidence for rule Shirt} \rightarrow \text{Trouser} = \frac{3}{6} \times 100 = 50\%$$

$$\text{Confidence for rule Trouser} \rightarrow \text{Shirt} = \frac{3}{5} \times 100 = 60\%$$

Only the following three association rules fulfil the confidence criteria of confidence $\geq 60\%$.

Tie $\rightarrow$ Coat; Trouser $\rightarrow$ Coat; Trouser $\rightarrow$ Shirt

Thus, the Arpiori algorithm is able to determine the frequent item sets, which can be used to determine the association rules. One of the major advantages of the Apriori algorithm is that it is a very easy algorithm to implement; however, it requires the transaction database to be memory resident.

14.9 APPLICATIONS OF DATA MINING

Some of the applications of data mining are as follows:

- **Marketing and sales data analysis:** A company can use customer transactions in their database to segment the customers into various types. Such companies may launch products for specific customer types.
- **Investment analysis:** Customers can look at the areas where they can get good returns by applying data mining.
- **Loan approval:** Companies can generate rules for giving loans depending upon the dataset they have. On that basis, they may decide to whom and what amount of loan should be given.
- **Fraud detection:** By finding the correlation between faults, new faults can be detected by applying data mining.
- **Network management:** By analysing patterns generated by data mining for the networks and their faults, the faults can be minimised as well as future hardware and software needs of the network can be predicted.
- **Brand Loyalty:** Given a customer and the product he/she uses, predict whether the customer will change their products.

Check Your Progress – 4

1) What is association rule mining?

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2) What are the applications of data mining in the banking domain?

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3) Apply the Apriori algorithm for generating frequent itemset in the following dataset:

Transaction ID	Items purchased
T _A	PI ₂ PI ₅
T _B	PI ₁ PI ₃
T _C	PI ₁ PI ₂ PI ₃ PI ₄
T _D	PI ₁ PI ₂ PI ₃

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14.10 SUMMARY

This unit first introduces the concepts of data warehousing systems. The data warehouse is a technology that collects operational data from several operational systems, refines it and stores it in its own multidimensional model, such as Star schema or Snowflake schema. The data of a data warehouse can be indexed and can be used for analyses through data mining. Data mining is the process of automatic extraction of interesting but not known information in large databases. Basic data mining tasks are Classification, Clustering and Association rules. The classification task maps data into predefined classes. Clustering task groups objects with similar properties/behaviour into the same group. Association rules find the association relationship among a set of objects. Data mining can be applied in many areas, whether it is Games, Marketing, Bioscience, Loan approval, Fraud detection etc.

Please go through further readings for more details on data warehousing and data mining.

14.11 SOLUTIONS/ANSWERS

Check Your Progress – 1

- 1) A Data Warehouse is a repository of processed but integrated information that can be used for queries and analysis. Data and information are extracted from heterogeneous sources. It is subject-oriented, time-variant integrated data, which is not changed once put in a data warehouse.
- 2)
 - Integrated data
 - Subject-oriented data
 - Time-variant data
 - Non-volatile
- 3) ETL is Extraction, transformation, and loading. ETL refers to the methods involved in accessing and manipulating data available in various sources and loading it into the target data warehouse. The following are some of the transformations that may be used during ETL:
 - Filter Transformation
 - Joiner Transformation
 - Aggregator transformation
 - Sorting transformation.

Check Your Progress – 2

- 1) A dimension may be equated with a reference object. For example, in a sales organisation, the dimensions may be *salesperson*, *product* and *period* of information. Each of these is a dimension. The fact table will represent the fact relating to the dimensions. For this example, a fact table will include *sales* (in rupees) made by a particular *salesperson* for a specific *product* for a certain *period*. Fact is actual data. A fact, thus, represents an aggregation of relational data on the dimensions.
- 2) The primary difference between them is that the Snowflake schema uses a normalised dimensional table.
- 3) Data mining is the process of automatic extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) information or patterns from large data stored in large databases or data warehouses.
- 4) Different data-mining tasks are - Classification, Clustering and Association Rule Mining.

Check Your Progress – 3

- 1) The classification task maps data into predefined groups or classes. Data are represented as tuples, which consist of a set of predicating attributes and a goal attribute. The task is to discover some kind of relationship between the predicating attributes and the goal attribute so that the discovered knowledge can be used to predict the class of new tuple(s).

Some examples of classification are:
Classification of students' grades depending on their marks in previous examinations.
Classification of customers as good or bad customers in a bank.
- 2) The task of clustering is to group the tuples with similar attribute values into the same class. Given a database of tuples and an integer value K , Clustering defines mapping, such that tuples are mapped to K different clusters.
- 3) In classification, the classes are predetermined, but in the case of clustering, the groups are not predetermined. Only the number of clusters is decided by the user.

Check Your Progress – 4

- 1) The basic idea of association rule mining is to find unknown and interesting associations among various data items.
- 2) Data mining applications in banking are as follows:
 - 1) Detecting patterns of fraudulent credit card use.
 - 2) Identifying good customers.
 - 3) Determining whether to issue a credit card to a person or not.
 - 4) Finding hidden correlations between different financial indicators.
- 3) The dataset D given for the problem is:

Transaction ID	Items purchased
T _A	PI ₂ PI ₅
T _B	PI ₁ PI ₃
T _C	PI ₁ PI ₂ PI ₃ PI ₄
T _D	PI ₁ PI ₂ PI ₃

Assuming the minimum support as 50% for calculating the large item sets. As we have 4 transactions, at least 2 transactions should have the data item.

First Scan:

C₁: PI₁:3, PI₂:3, PI₃:3, PI₄:1, PI₅:1

L₁: PI₁:3, PI₂:3, PI₃:3

C₂: PI₁PI₂, PI₁PI₃, PI₂PI₃

Second Scan:

C₂: PI₁PI₂:2, PI₁PI₃:3, PI₂PI₃:2

L₂: PI₁PI₂:2, PI₁PI₃:3, PI₂PI₃:2

C₃: PI₁PI₂PI₃

Pruned C₃: PI₁PI₂PI₃

Third scan

L₃: PI₁PI₂PI₃: 2

Frequent item sets L={L₁, L₂, L₃}

14.12 FURTHER READINGS

- 1) *Data Mining Concepts and Techniques*, J Han, M Kamber, Morgan Kaufmann Publishers, 2001.
- 2) *Data Mining*, A K Pujari, 2004.